

ShopFlow AWS Project

ECS Fargate Design

Containerised Workloads - Task Definitions - Rolling Deploys

Module 11 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
Author	Manish Kumar Sharma
Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon ECS with Fargate runs ShopVerse as a containerised service without managing any EC2 instances. You define the container spec (Task Definition), and AWS provisions and manages the underlying compute. ECS Service ensures the desired number of tasks is always running and performs rolling updates on new deploys.

- Fargate: serverless containers -- no EC2 fleet to patch or right-size
- Each task gets its own ENI with a private IP in the VPC
- Secrets injected from Secrets Manager and SSM at task startup
- DeploymentCircuitBreaker: auto-rollback if new tasks fail health checks
- Container Insights enabled for per-service CloudWatch metrics

2. Task Definition

Setting	Value	Notes
Family	shopflow-shopverse-dev	Versioned on each update
Network Mode	awsvpc	Each task gets own ENI + private IP
Launch Type	FARGATE	Serverless -- no EC2 management
CPU	512 units (0.5 vCPU)	Minimum for Spring Boot
Memory	1024 MB (1 GB)	JVM Xmx512m + 512m overhead
Container Port	8080	Spring Boot server port
Health Check	curl /actuator/health	60s start period -- Spring Boot init time
Log Driver	awslogs	Streams stdout/stderr to CloudWatch Logs

3. Environment Variable Injection

Env Variable	Source	Secret/Parameter Path
DB_HOST	SSM	/shopflow/dev/db/endpoint
DB_PORT	SSM	/shopflow/dev/db/port
DB_NAME	SSM	/shopflow/dev/db/name
REDIS_HOST	SSM	/shopflow/dev/redis/endpoint
DB_USER	Secrets Manager	/shopflow/db/app -- username
DB_PASS	Secrets Manager	/shopflow/db/app -- password
JWT_SECRET	Secrets Manager	/shopflow/app/jwt-secret -- jwt_secret
REDIS_AUTH_TOKEN	Secrets Manager	/shopflow/app/redis-auth -- auth_token

4. ECS Service

Setting	Value	Reason
Desired Count	1 (dev)	Single task for learning; increase for prod
Min Healthy Percent	100%	New task healthy before old stops
Max Percent	200%	Allows 2 tasks during rolling deploy
Circuit Breaker	Enabled + Rollback	Auto-rollback on failed health checks
ALB Integration	shopflow-tg-arn	Registers task IPs with Target Group

5. EC2 vs ECS Fargate

Attribute	EC2 (Module 8)	ECS Fargate (Module 11)
Management	OS + JVM + systemd	Docker image only
Deployment	Copy JAR + restart	Push image tag, ECS rolls
Scaling	Manual / ASG	Change DesiredCount
Cost model	Pay 24/7 for instance	Pay per task per second
Best for	Steady predictable load	Variable containerised workloads